Cat Crusher

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Cat Crusher is a distortion, filter and compressor in one plugin. It makes drums, bass and leads thicker, grittier and louder: choose one of ten distortion types, add a hard digital crunch with Mech, shape the tone with a resonant 24 dB filter (low-pass, high-pass or band-pass), then glue everything with a one-knob compressor with automatic make-up gain, so the level stays the same at any Amount and only the density changes. A hidden soft clipper and safety limiter keep peaks under control. Built for dance, house, UK hardcore and hardstyle.

Quick start

• Pick the preset closest to your sound (Fat Bass, Hardstyle Kick Crunch, UK Hardcore Stab, Gritty Drums).
• Set Type first, then raise Drive until it sounds right; the Distortion curve shows the shape you are using.
• Use Cutoff to remove harsh highs (Filter type Low-pass), thin the lows (High-pass) or make a telephone or radio sound (Band-pass), Amount for density, and Mix for parallel crush.
• Compare with the plugin bypassed and set Volume so both are equally loud.
• The round gear button in the header opens Settings: the plugin version, a link to the GitHub page and the releases page (it opens your browser; the plugin itself never goes online), and a switch for the info bubbles. Hover any control for about one second to see what it does; the round i button also turns the bubbles on or off.

How the window works

• Each knob sits directly under the display it changes; the displays react to the knobs and the knobs react to the displays.
• Meters show the real signal in dB with a dB scale: input and output on the far right.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• The Limiter switch in the header adds a safety limiter at the output (useful while experimenting).
• Presets are in the header; the dice picks random settings (some controls are excluded on purpose).

Controls

Distortion

Distortion	Choice: Off / On (default On)	Distortion on/off.
Type	Choice: Tube, Soft Sat, Transistor, Tape, Diode, Hard Clip, Fuzz, RAT, Overdrive, Rectifier (default Tube)	Distortion type. Tube: warm even harmonics. Soft Sat: smooth and symmetric, great on buses. Transistor: console-style, punchy and forward. Tape: gentle and rounded. Diode: edgy and asymmetric. Hard Clip: tight, loud and aggressive. Fuzz: thick, gated and woolly. RAT: hard-clipped, gritty distortion pedal style. Overdrive: smooth, mid-forward tube-pedal style. Rectifier: adds a bright octave-ish edge.
Drive	0 to 100 % default 30 %	Drive: how hard the selected distortion type is pushed. Higher = more saturation and harmonics.
Mech	0 to 100 % default 0 %	Mech: harder, more aggressive digital-style crunch (folds and clips the wave). 0 = off.

Filter

Filter type	Choice: Low-pass, High-pass, Band-pass (default Low-pass)	Filter type: Low-pass keeps the lows and cuts the highs, High-pass keeps the highs and cuts the lows, Band-pass keeps only a band around the cutoff.
Filter	Choice: Off / On (default On)	Filter on/off.
Cutoff	40 to 20000 Hz default 20000 Hz	Cutoff: frequency of the 24 dB filter (low-pass, high-pass or band-pass). As a low-pass it rolls off the harsh highs that distortion creates.
Res	0 to 100 % default 10 %	Resonance: boosts the frequencies right at the cutoff for a vocal or squelchy edge.

Compressor

Compressor	Choice: Off / On (default On)	Compressor on/off.
Phat mode	Choice: Off / On (default Off)	Phat mode: adds low-end weight before the compressor, a slower release and a soft ceiling. Great on kicks and bass.
Amount	0 to 100 % default 30 %	Compressor amount: lowers the threshold and raises the ratio together. Make-up gain is automatic, so the level stays the same at any amount and only the density changes.

Master

Volume	-24 to 12 dB default 0 dB	Volume: final output level. Match it to the bypassed level when comparing.
Mix	0 to 100 % default 100 %	Mix: dry/wet blend. Lower values keep the original punch (parallel crush).

Recommendations

- Kick: Type Diode or Transistor, Drive 60 to 80, Phat mode on, Mix 80 to 90 to keep the click.
- Bass: Type Transistor, Drive 50, Cutoff 4 kHz, Phat mode on; keep Mech low so the sub stays clean.
- Radio or telephone: Filter type Band-pass, Cutoff 1 to 3 kHz, Resonance 30 to 60, Mech 40.
- Lead or stab: Mech 30 to 60 and Resonance 30 to 50 for screech; roll Cutoff down if it gets harsh.
- Drum bus: Type Tape, Drive 20 to 30, Amount 25 to 40, Mix 100 for glue or 60 to 70 for parallel punch.
- If it sounds louder but worse, lower Volume: loudness fools the ear.

Presets

Init, Warm Bus, Gritty Drums, Fat Bass, Radio Crush, Screech Lead, Hardstyle Kick Crunch, UK Hardcore Stab, Smooth Lows, Telephone Crunch, Mech Mayhem

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The picture is a screenshot of the real compiled plugin window.



Signal flow

Input > Distortion > Filter > Compressor > Master (volume, dry/wet mix) > output. Each of the first three stages has its own On button. The Nasty range and Limiter switches in the header act at the very end (Nasty range widens the knob ranges, Limiter is a safety limiter on the output).

Header bar

1	**Preset menu**	Factory presets (our own). Use the arrows to step through them.
2	**Randomize**	Set the controls to random values (output level and on/off switches are kept).
3	**Nasty range**	Nasty range: lets Drive, Fuzz, Squeeze, Warm and the other heavy knobs go far past 100 % for really nasty, OTT and Vengeance style sounds. Off keeps every knob at its normal top.
4	**Limiter**	Safety limiter on the output: keeps the level from ever passing 0 dBFS, smooth and transparent. Turn it off if you want the raw output (for example to let your own limiter do the work).
5	**Settings**	Settings: version, link to the GitHub page for new versions
6	**Info bubbles (i)**	Info bubbles: hover a control for one second to see what it does. Click to turn them off or on.

A. Distortion



Display A: Transfer curve of the distortion. Horizontal axis: input level from -1 to +1. Vertical axis: output level. The orange line is the shape the type and Drive give; a flatter top means harder clipping. The white dot is the live input level, so you see where the signal sits on the curve. The label top right shows the gain applied by Drive.

7	**Drive**	Drive: how hard the selected distortion type is pushed. Higher = more saturation and harmonics. (shown: 30 %)
8	**Mech**	Mech: harder, more aggressive digital-style crunch (folds and clips the wave). 0 = off. (shown: 0 %)
9	**On**	Distortion on/off.
10	**Distortion type**	Distortion type. Tube: warm even harmonics. Soft Sat: smooth and symmetric, great on buses. Transistor: console-style, punchy and forward. Tape: gentle and rounded. Diode: edgy and asymmetric. Hard Clip: tight, loud and aggressive. Fuzz: thick, gated and woolly. RAT: hard-clipped, gritty distortion pedal style. Overdrive: smooth, mid-forward tube-pedal style. Rectifier: adds a bright octave-ish edge. (shown: Tube)

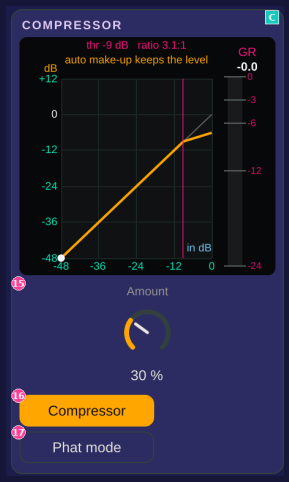
B. Filter



Display B: Frequency response of the 24 dB filter. Horizontal axis: frequency (log, 50 Hz to 20 kHz). Vertical axis: gain in dB (+12 to -36). The teal line follows Cutoff, Res and the filter type; the label shows the cutoff.

11	**Cutoff**	Cutoff: frequency of the 24 dB filter (low-pass, high-pass or band-pass). As a low-pass it rolls off the harsh highs that distortion creates. (shown: 20000 Hz)
12	**Res**	Resonance: boosts the frequencies right at the cutoff for a vocal or squelchy edge. (shown: 10 %)
13	**Filter type**	Filter type: Low-pass keeps the lows and cuts the highs, High-pass keeps the highs and cuts the lows, Band-pass keeps only a band around the cutoff. (shown: Low-pass)
14	**On**	Filter on/off.

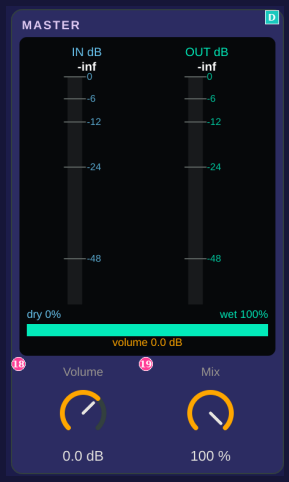
C. Compressor



Display C: Compressor curve plus gain reduction meter. Horizontal axis: input level in dB, vertical axis: output level in dB; the thin diagonal is 'no change'. The orange line bends at the threshold (pink vertical line); the labels give threshold, ratio and make-up gain, all set together by Amount. The white dot is the live level. The bar on the right (GR) shows how many dB the compressor is pulling down right now.

15	**Amount**	Compressor amount: lowers the threshold and raises the ratio together. Make-up gain is automatic, so the level stays the same at any amount and only the density changes. (shown: 30 %)
16	**On**	Compressor on/off.
17	**Phat mode**	Phat mode: adds low-end weight before the compressor, a slower release and a soft ceiling. Great on kicks and bass.

D. Master



Display D: Input and output meters in dB (peak, with a hold line) and a dry/wet bar: pink is dry, teal is wet. Volume is printed under the bar.

18	**Volume**	Volume: final output level. Match it to the bypassed level when comparing. (shown: 0.0 dB)
19	**Mix**	Mix: dry/wet blend. Lower values keep the original punch (parallel crush). (shown: 100 %)

Under the hood

• Distortion has ten types. Drive goes to 300 % and Mech to 200 % in the extended range (hold the knob past the normal range in the plugin; the test page shows the same ranges).
• Filter is a 24 dB per octave design with three modes. Band-pass is scaled so it has unity gain at the cutoff, so switching types does not jump in level.
• Compressor Amount moves threshold, ratio and make-up gain together. The make-up is automatic: a slow 400 ms RMS follower compares what went in with what comes out and keeps the loudness, with the gain rising over about 50 ms and falling over about 300 ms so it never pumps audibly by itself.
• Phat mode adds low-end weight before the compressor, uses a slower release and a soft ceiling: made for kicks and bass.
• Knob changes reach the sound within one audio buffer; smoothing never exceeds 20 ms so there is no zipper noise and no sluggish feel.
• The output has two safety stages: a hidden soft clipper (no knob) that shaves only the tallest peaks, then the Limiter switch in the header, which stops anything passing the ceiling. Nasty range in the header widens the knob ranges.

Install and use

• Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
• Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
• Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.